

Sparse Cell Search Method for Centralized Target Detection in Multistatic Radar

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The cell search problem in centralized target detection based on spatial resolution cell (SRC) is investigated. In centralized target detection, the SRC is utilized to register observation data from different radar nodes. Subsequently, each SRC within the common surveillance region is detected to determine the presence of a target. This process is referred to as cell search. However, traversal detection of all SRCs causes great computational complexity, making it challenging to apply in practice. A sparse cell search method is proposed to solve the problem. Data registration based on SRCs may result in adjacent SRCs sharing the same signal response of a radar node's range resolution cell. This symbiosis effect causes false alarms in the SRCs without a target. Such false alarm cells are referred to as symbiosis cells. The sparse cell search method can utilize the spatial dependency between the target and symbiosis cells to achieve rapid target search. The proposed method significantly reduces the computational complexity of the cell search, improves the detection efficiency for the target, and has high engineering application value. Comparative experiments with the traversal search method further validate the effectiveness of the proposed method.

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I. INTRODUCTION

Multistatic radar system consists of multiple spatially separated radar nodes [1]. The flexible position and working modes of the radar nodes enhance the survivability of the multistatic radar [2], [3], [4], the different observing angles increase the quantity and dimensionality of the collected information of the target [5], [6], [7], and the fusion processing of data from radar nodes further improve the detection performance [8], [9], [10], [11], [12]. Multistatic radar systems have attracted increasing attention in recent literature due to their outstanding performance.

The multistatic radar can be divided into two groups according to the data fusion strategy: distributed multistatic radar system and centralized multistatic radar system. In distributed multistatic radar system, each radar node independently processes data [13], [14], [15]. The local decisions are transmitted to a fusion center for joint processing. The advantages of distributed target detection include lower communication bandwidth requirements and more flexible scalability. However, the independent data processing by each radar node may result in discrepancies in target detection results and information isolation, ultimately leading to diminished detection performance. In contrast, the centralized multistatic radar system transmits the observed data from all radar nodes to the fusion center for comprehensive processing. This fusion framework enables the full utilization of target information within the raw data [16], minimizes information loss and inconsistency, and facilitates more effective data fusion, thereby significantly enhancing the target detection performance [17].

While centralized target detection can efficiently process data from all radar nodes, it requires precise matching of echo signals across different radar nodes, known as data registration [20]. Correct data registration is a prerequisite for centralized target detection. The echo signals of the same target exhibit different time delays and lengths in different channels due to the varying spatial positions, observation angles, and resolutions of each radar node, thereby complicating the data registration process. Utilizing spatial resolution cell (SRC) for data registration is a feasible solution. There is extensive research on the multistatic radar's SRC. An SRC-based centralized target detection methodology was introduced in [8]. Data registration is performed based on the SRC, where the SRC is defined as the intersection of all radar nodes' range resolution cells (RRCs). The common surveillance region (CSR) of the MIMO radar is partitioned into uniformly shaped spatial grid resolution cells in [18]. Based on the grid cells, two data registration methods, the center-method and the comparison-method, were proposed. The problem of centralized target detection with registration errors was researched in [19] and [20]. Zhang et al. [9] derived the relationship between the size of SRC and the detection performance of multistatic radar and designed an elliptical SRC that can control the lower threshold of target detection performance. In this architecture, the SRC is the basic unit of centralized target detection. Dense SRCs are needed to cover the entire detection space. Then, the

multistatic system employs a cell search method to detect all SRCs to determine the target's position. This step determines the computational complexity of the centralized detection and the detection efficiency for the target. We expect to spend less computing resources and locate targets faster in dense SRCs. Regrettably, research on cell search methods remains limited.

In the open literature, the research on the SRC focuses on the following aspects: design for SRC, detector design for interested SRC, and impact of registration error on detection performance. For the cell search method, the literature only mentions traversing all SRCs in the CSR to determine the target's position. While the traversal search method has a certain application value, it will cause great computational complexity when dealing with dense SRC. Furthermore, an inappropriate search starting point can significantly waste computational resources, which seriously reduces the detection efficiency for the target. Consequently, optimizing the cell search method is crucial for improving target detection efficiency. Interestingly, in the process of studying SRC, we observed the symbiosis effect introduced by data registration based on the SRC, which provides a new idea for optimizing the cell search method.

Since the SRC is considerably smaller than the radar node's RRC, during the data registration, adjacent SRCs may share the signal response of the radar nodes' RRCs containing the target. This causes nontarget SRCs to exceed the detection threshold with a certain probability. This phenomenon is referred to as the symbiosis effect. These nontarget SRCs that exceed the detection threshold are referred to as symbiosis cells. So far, only a few scholars have studied this. It was observed in [21] that the intersection of different transmit–receive channels can lead to ghost targets. The article proposed a two-step detection procedure based on grid space searching to eliminate these ghost targets. The symbiosis relationship of the adjacent SRCs was studied in [8]. A peak search method was proposed to determine the position of the target. This method is only applicable to the single target scenario. The authors in [22] and [23] investigated the multitarget detection problem in MIMO radar. Affected by the symbiosis effect, many false targets will be generated at the intersection of the RRCs where multiple targets are located. The centralized detection in a deception jamming scenario was discussed in [2]. Deception jamming complicates the impact of symbiosis effect. A deception dominated SRC discrimination and interference cancellation algorithm was proposed to solve the problem. These scholars are dedicated to studying methods to mitigate the negative impact of the symbiosis effect. While the symbiosis cells obstruct the target detection, they indicate the general area of the target. So long as a symbiosis cell is found, the target can be located by the position relation between symbiosis cells and the target. Compared with searching for a single target SRC, searching for many symbiosis cells that are continuously distributed in space is obviously much simpler.

This article proposes a sparse cell search method for centralized target detection based on SRC. In centralized target

detection, the SRCs are located in the same transmit–receive channel as the target will share the energy of the target echo, thereby producing a symbiosis effect. Consequently, the symbiosis cells introduced by the symbiosis effect are highly related to the target in space. They are continuously distributed within the RRCs of different radar nodes covering the target. The SRC is divided at a certain distance interval in the CSR, which can reduce the number of SRCs that need to be detected. The order of detection is determined according to the multistatic radar's geometric configuration and the position of the CSR. When any symbiosis cell is detected, we can utilize this SRC as a starting point to locate the target by the position relation. The proposed method can significantly reduce the computational complexity while ensuring the detection probability for the target. Furthermore, an alternating sparse cell search method is proposed to handle the problem of discontinuous spatial distribution of symbiosis cells under low signal-to-noise ratio (SNR) conditions. Numerical simulation experiments demonstrate the efficiency of the proposed sparse cell search method and reveal its significant potential for practical applications.

The rest of this article is organized as follows. Section II derives the generalized likelihood ratio test (GLRT) detector for multistatic radar and elucidates the model of the symbiosis effect. Section III introduces the principle of the sparse cell search method and provides the implementation steps of the method. Section IV presents the results of experimental simulations. Finally, Section V concludes this article.

II. SIGNAL MODEL

Radar nodes differ in spatial positions, observed angles, and range resolutions, making data registration across different nodes challenging. Registering the observed data based on SRC within a unified coordinate system is feasible. However, because of the inconsistent size of the multistatic radar's SRC and the monostatic radar's RRC, this registration method will cause the symbiosis effect. In this section, we establish the signal model of the multistatic radar and derive the centralized target detector based on the GLRT criterion. The mechanism of the symbiosis effect is analyzed, and the model of the symbiosis cell is provided.

A. Signal Model of Multistatic Radar

Consider a multistatic radar system featuring N radars, as illustrated in Fig. 1. The position of the n th radar node is denoted by $\mathbf{p}_{R,n}$, $\mathbf{p}_{R,n} = [x_{R,n}, y_{R,n}]^T$. There is a moving target in the CSR. The position of the target is denoted by $\mathbf{p} = [x, y]^T$, and the velocity of the target is $\mathbf{v} = [v_x, v_y]^T$. All radars observe the target simultaneously. Each radar only receives the signal transmitted by itself [24].

Suppose that the n th radar node transmits M_n pulse signals, then the received m th pulse echo signal can be expressed as

$$r_n(t) = \alpha_n s_n(t - \tau_n) e^{j2\pi f_{d,n}(m-1)T_n} + w_{n,m}(t) \quad (1)$$

$$\begin{cases} \tau_n = \tau_n(\mathbf{p}) = \frac{2}{c} \|\mathbf{p} - \mathbf{p}_{R,n}\| \\ f_{d,n} = f_{d,n}(\mathbf{p}, \mathbf{v}) = \frac{2(\mathbf{p} - \mathbf{p}_{R,n})^T \mathbf{v}}{\lambda_n \|\mathbf{p} - \mathbf{p}_{R,n}\|} \end{cases} \quad (2)$$

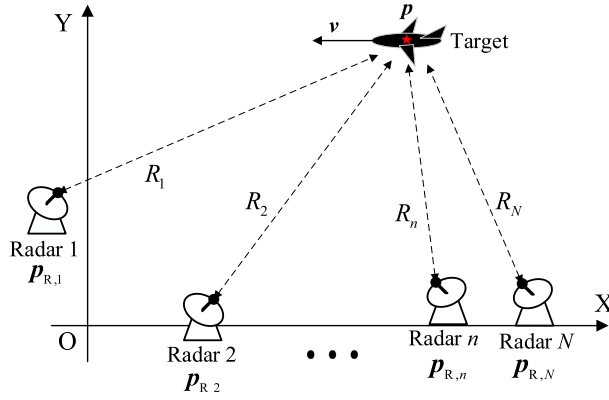


Fig. 1. Detection scenario of multistatic radar system.

where α_n is the complex amplitude of the signal, which is assumed to be an unknown deterministic parameter [25], [26], [27], τ_n is the time delay of the signal, $f_{d,n}$ is the Doppler frequency shift term, T_n is the repetition period, $w_{n,m}(t)$ is the noise modeled as a complex Gaussian random process, c is the speed of light, and λ_n is the wavelength.

After discrete sampling, the observed data of the n th radar node can be written as

$$\mathbf{Z}_n = \alpha_n \mathbf{d}_n(\mathbf{p}, \mathbf{v}) s_n(\mathbf{p}) + \mathbf{W}_n \quad (3)$$

where $\mathbf{d}_n(\mathbf{p}, \mathbf{v}) = [1, \dots, e^{j2\pi f_{d,n}(m-1)T_n}, \dots, e^{j2\pi f_{d,n}(M_n-1)T_n}]^T \in \mathbb{C}^{M_n \times 1}$, $\mathbf{s}_n(\mathbf{p}) = [s_{n,1}(\mathbf{p}), s_{n,2}(\mathbf{p}), \dots, s_{n,L_n}(\mathbf{p})] \in \mathbb{C}^{1 \times L_n}$ represents L_n snapshot signals, $\mathbf{W}_n = [\mathbf{w}_{n,1}, \dots, \mathbf{w}_{n,m}, \dots, \mathbf{w}_{n,M_n}]^T \in \mathbb{C}^{M_n \times L_n}$. The noise samples from different radar node are independent. They are zero-mean complex Gaussian valuables with

$$\mathbb{E} \{ \mathbf{w}_{n,l} \mathbf{w}_{n',l'}^H \} = \begin{cases} \mathbf{C}_n & n = n' \\ \mathbf{0} & \text{others.} \end{cases} \quad (4)$$

The target detection problem can be transformed into a binary hypothesis testing problem [28]

$$\begin{aligned} H_0 : & \begin{cases} \mathbf{Z}_n = \mathbf{W}_n \\ \tilde{\mathbf{Z}}_n = \tilde{\mathbf{W}}_n \end{cases} \\ H_1 : & \begin{cases} \mathbf{Z}_n = \alpha_n \mathbf{d}_n(\mathbf{p}, \mathbf{v}) s_n(\mathbf{p}) + \mathbf{W}_n \\ \tilde{\mathbf{Z}}_n = \tilde{\mathbf{W}}_n \end{cases} \end{aligned} \quad (5)$$

where $\tilde{\mathbf{Z}}_n = [\tilde{z}_{n,1}, \tilde{z}_{n,2}, \dots, \tilde{z}_{n,K_n}] \in \mathbb{C}^{M_n \times K_n}$, $K_n > M_n$ represents K_n range cells around the cell under test (CUT), which is selected as reference cells to estimate the spectral characteristics of noise. H_0 is the hypothesis that the CUT contains only noise components, and H_1 is the hypothesis that the SRC contains both the desired target and noise components.

Under H_0 , the probability density function (PDF) of the n th radar's observations is

$$\begin{aligned} f_n(\mathbf{Z}_n, \tilde{\mathbf{Z}}_n | \mathbf{C}_n, H_0) &= \prod_{l=1}^{L_n} \frac{1}{\pi^{M_n} \det(\mathbf{C}_n)} e^{-\mathbf{z}_{n,l}^H \mathbf{C}_n^{-1} \mathbf{z}_{n,l}} \prod_{k=1}^{K_n} \frac{1}{\pi^{M_n} \det(\mathbf{C}_n)} e^{-\tilde{\mathbf{z}}_{n,k}^H \mathbf{C}_n^{-1} \tilde{\mathbf{z}}_{n,k}}, \quad (6) \\ &= \left(\frac{1}{\pi^{M_n} \det(\mathbf{C}_n)} e^{-\text{tr}(\mathbf{C}_n^{-1} \mathbf{T}_n^0)} \right)^{L_n + K_n} \end{aligned}$$

where $\mathbf{T}_n^0$ is defined as

$$\mathbf{T}_n^0 = \frac{\sum_{l=1}^{L_n} \mathbf{z}_{n,l} \mathbf{z}_{n,l}^H + \sum_{k=1}^{K_n} \tilde{\mathbf{z}}_{n,k} \tilde{\mathbf{z}}_{n,k}^H}{L_n + K_n}. \quad (7)$$

Similarly, the PDF of the n th radar's observations under H_1 can be expressed as

$$\begin{aligned} f_n(\mathbf{Z}_n, \tilde{\mathbf{Z}}_n | \alpha_n, \mathbf{C}_n, \mathbf{p}, \mathbf{v}, H_1) &= \left(\frac{1}{\pi^{M_n} \det(\mathbf{C}_n)} e^{-\text{tr}(\mathbf{C}_n^{-1} \mathbf{T}_n^1)} \right)^{L_n + K_n} \end{aligned} \quad (8)$$

where $\mathbf{T}_n^1 = \frac{\sum_{l=1}^{L_n} \hat{\mathbf{z}}_{n,l} \hat{\mathbf{z}}_{n,l}^H + \sum_{k=1}^{K_n} \tilde{\mathbf{z}}_{n,k} \tilde{\mathbf{z}}_{n,k}^H}{L_n + K_n}$, $\hat{\mathbf{z}}_{n,l} = \mathbf{z}_{n,l} - \alpha_n s_{n,l}(\mathbf{p}) \mathbf{d}_n(\mathbf{p}, \mathbf{v})$.

The GLRT criterion is applied to the binary hypothesis testing problem [29], [30]. Assuming that the signals received by different radar nodes are independent, the joint detection statistic of the multistatic radar is

$$\frac{\max_{\alpha_1, \dots, \alpha_N, \mathbf{C}_1, \dots, \mathbf{C}_N, \mathbf{p}, \mathbf{v}} \prod_{n=1}^N f_n(\mathbf{Z}_n, \tilde{\mathbf{Z}}_n | \alpha_n, \mathbf{C}_n, \mathbf{p}, \mathbf{v}, H_1)}{\max_{\mathbf{C}_1, \dots, \mathbf{C}_N} \prod_{n=1}^N f_n(\mathbf{Z}_n, \tilde{\mathbf{Z}}_n | \mathbf{C}_n, H_0)} \underset{H_0}{\overset{H_1}{\geq}} \eta \quad (9)$$

where η is the test threshold determined by the false alarm probability.

The maximum likelihood estimation of the covariance matrix is the sample covariance matrix [31], [32], [33]. Consequently, the statistic can be maximized on the sample covariance matrix $\mathbf{C}_n$. Equation (9) can be simplified as

$$\frac{\prod_{n=1}^N \det(\mathbf{T}_n^0)^{L_n + K_n}}{\min_{\alpha_1, \dots, \alpha_N, \mathbf{p}, \mathbf{v}} \prod_{n=1}^N \det(\mathbf{T}_n^1)^{L_n + K_n}} \underset{H_0}{\overset{H_1}{\geq}} \eta. \quad (10)$$

To simplify the operation, we define the matrix $\mathbf{V}_n = \sum_{k=1}^{K_n} \tilde{\mathbf{z}}_{n,k} \tilde{\mathbf{z}}_{n,k}^H \in \mathbb{C}^{M_n \times M_n}$, where $K_n > M_n$. Only under this condition is the matrix $\mathbf{V}_n$ a nonsingular matrix. Moreover, $\mathbf{V}_n$ is a positive definite Hermitian matrix. Equation (10) can be simplified as

$$\frac{\prod_{n=1}^N \left(1 + \sum_{l=1}^{L_n} \mathbf{z}_{n,l}^H \mathbf{V}_n^{-1} \mathbf{z}_{n,l} \right)^{L_n + K_n}}{\min_{\alpha_1, \dots, \alpha_N, \mathbf{p}, \mathbf{v}} \prod_{n=1}^N \left(1 + \sum_{l=1}^{L_n} \hat{\mathbf{z}}_{n,l}^H \mathbf{V}_n^{-1} \hat{\mathbf{z}}_{n,l} \right)^{L_n + K_n}} \underset{H_0}{\overset{H_1}{\geq}} \eta. \quad (11)$$

Then, minimize the denominator of (11) on $\alpha_1, \dots, \alpha_N$, and we can get [34]

$$\max_{\mathbf{p}, \mathbf{v}} \prod_{n=1}^N \left(1 - \frac{q_n(\mathbf{p}, \mathbf{v})}{p_n} \right)^{-(L_n + K_n)} \underset{H_0}{\overset{H_1}{\geq}} \eta \quad (12)$$

where

$$p_n = 1 + \sum_{l=1}^{L_n} \mathbf{z}_{n,l}^H \mathbf{V}_n^{-1} \mathbf{z}_{n,l} \quad (13)$$

$$q_n(\mathbf{p}, \mathbf{v}) = \frac{\left| \sum_{l=1}^{L_n} s_{n,l}^*(\mathbf{p}) \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{z}_{n,l} \right|^2}{\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})}. \quad (14)$$

The derivation can be seen in Appendix A.

After a logarithm transform of (14), the GLRT detector for multistatic radar can be expressed as

$$\psi = \max_{\mathbf{p}, \mathbf{v}} - \sum_{n=1}^N (L_n + K_n) \ln \left(1 - \frac{q_n(\mathbf{p}, \mathbf{v})}{p_n} \right) \stackrel{H_1}{\geq} \eta. \quad (15)$$

A universal threshold can be set in (15) to satisfy the constant false alarm constraint for any given desired false alarm probability. The threshold only depends on the parameters of the multistatic radar system and can be obtained by Monte Carlo experiments. Through the GLRT detector, we can decide whether there is a target in an SRC.

B. Symbiosis Effect Model

The SRC is the basis of data registration and the basic processing unit of the detector. The SRC of the multistatic radar system is the intersection of the radar nodes' RRCs. Consequently, the shape and size of the SRC are intrinsically associated with the topology of the multistatic radar system. In general, the SRCs show a range of irregular geometries, complicating their division within the detection space. Numerous studies have explored the design of SRC, including grid cell, rhombic SRC, and elliptical SRC. The method proposed in this article works for SRC of any shape. For clarity, the CSR of the multistatic radar system is divided into identically shaped grid cells in a two-dimensional (2-D) rectangular coordinate system in this article.

The CSR of the multistatic radar is denoted as Γ , and it is divided into G spatial grid cells. Assuming that there is no registration error between radar nodes. The g th SRC is expressed as $\gamma(\mathbf{p}_{c,g})$. $\mathbf{p}_{c,g}(\mathbf{p}_{c,g} = [x_{c,g}, y_{c,g}]^T)$ represents the center coordinate of the SRC.

$\gamma(\mathbf{p}_{c,g})$ is defined as

$$\gamma(\mathbf{p}_{c,g}) = \left\{ (x, y) \mid -\frac{S_x}{2} \leq x - x_{c,g} < \frac{S_x}{2}, -\frac{S_y}{2} \leq y - y_{c,g} < \frac{S_y}{2}, (x, y) \in \Gamma \right\} \quad (16)$$

where S_x and S_y represent the dimensions of the grid cells along the x -axis and y -axis, respectively.

The radar's RRCs are typically represented as a ring centered on the radar. The inner and outer diameters of the ring are determined by the radial distance and range resolution, and the angular extent is determined by the angular resolution. The i th RRC of the n th radar node within the CSR is denoted as $\Theta_{n,i}$. $\Theta_{n,i}$ is defined as

$$\Theta_{n,i} = \left\{ (r, \theta) \mid -\frac{c}{4B_n} \leq r - r_{n,i} < \frac{c}{4B_n}, -\frac{\Delta\theta_n}{2} \leq \theta - \theta_{n,i} < \frac{\Delta\theta_n}{2}, (r, \theta) \in \Gamma \right\} \quad (17)$$

Common surveillance region

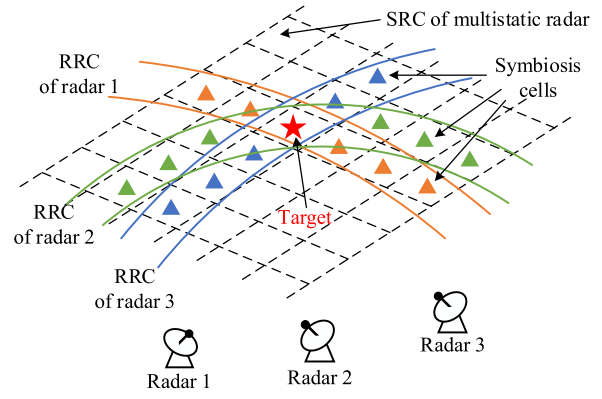


Fig. 2. Symbiosis effect caused by SRC-based data registration.

where B_n is the signal bandwidth, $r_{n,i}$ is the radial distance from the n th radar node to the center of the i th RRC, and $\theta_{n,i}$ and $\Delta\theta_n$ are the beam pointing direction and beamwidth of the n th radar node, respectively.

The mapping relationship between SRC and RRC can be expressed as

$$f_n : \gamma(\mathbf{p}_{c,g}) \rightarrow \Theta_{n,i} \quad (18)$$

$$\frac{c}{4B_n} \leq \|\mathbf{p}_{c,g} - \mathbf{p}_{R,n}\| - r_{n,i} < \frac{c}{4B_n} \quad (19)$$

$$g = 1, 2, \dots, G' < G$$

where $\mathbf{p}_{R,n}$ is the position of the n th radar node. All the G' SRCs that satisfy the constraint conditions described in (19) correspond to the $\Theta_{n,i}$.

Consequently, when a target is present in $\Theta_{n,i}$, the G' SRCs will match the target's echo with other radar nodes. However, the target is only located in one of these SRCs. The remaining $G' - 1$ SRCs without a target will also share the target's signal response.

As shown in Fig. 2, the symbiosis effect occurs in the SRCs with triangular markers. The different colors represent the shared energy coming from different radar nodes. For example, the SRCs with blue triangle markers share the signal response of radar 3 to the target. The symbiosis effect is a data misregistration caused by the inconsistent size of the multistatic radar's SRC and the monostatic radar's RRC. Therefore, the symbiosis cells are adjacent to the target cell in space. Moreover, when the SNR is sufficient, the symbiosis cells will be continuously distributed in space. Compared with searching for a single target SRC, it is obviously simple to search for a lot of symbiosis cells that are continuously distributed in space. So long as a symbiosis cell is found, the target can be located by the position relation between symbiosis cells and the target.

III. CELL SEARCH PROBLEM IN CENTRALIZED TARGET DETECTION

For the cell search method, target detection and localization performance depend on the size of the spatial grid cell. We must search for the optimal solution with dense

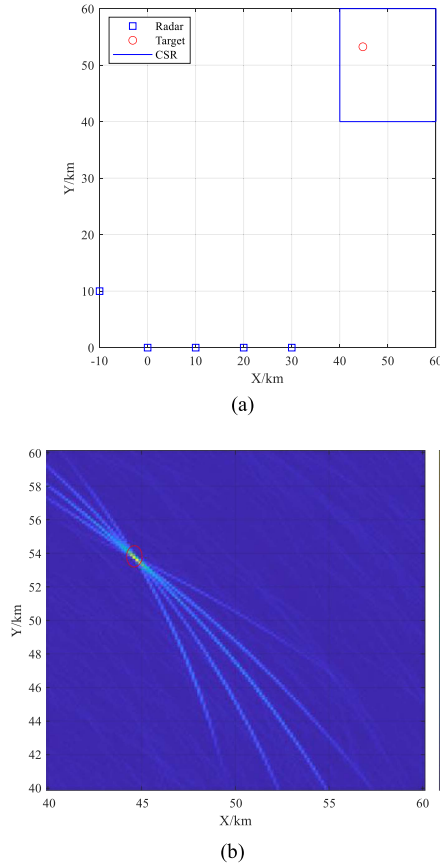


Fig. 3. Symbiosis cells in centralized target detection. (a) Geometrical configuration of multistatic radar. (b) Detection results of all grid cells in the CSR.

grids. In this case, the traversal search method causes great computing complexity, which is challenging in practice applications. This section investigates the cell search problem to address the conflict between detection performance and computing complexity.

In centralized target detection, the symbiosis cells are accompanied by the target. Fig. 3 shows the symbiosis cells in centralized target detection. The positions of the radar nodes are $(-10, 10)$, $(0, 0)$, $(10, 0)$, $(20, 0)$, and $(30, 0)$ km, respectively. The position and velocity of the target are $(44.8, 53.25)$ km and $(-150, -200)$ m/s. The CSR of the multistatic radar is a rectangle whose coordinates of the four vertices are $(40, 40)$, $(40, 60)$, $(60, 60)$, and $(60, 40)$ km. The size of the radar node's range cell is 150 m. The size of the multistatic radar's spatial grid cell is $150 \text{ m} \times 150 \text{ m}$. Considering the RCS flickering of the target, the SNRs of radar nodes are set to 16.5, 12, 18, 15, and 16.5 dB. Fig. 3(b) illustrates the traversal search results. The target is located at the red ellipse mark. The value of the detection statistic of the target cell is the largest. The symbiosis cells are continuously distributed across a large area within the CSR. They show a significant spatial correlation with the target cell and are continuously distributed within the RRCs of different radar nodes covering the target. This phenomenon provides the foundation for the proposed sparse cell search method.

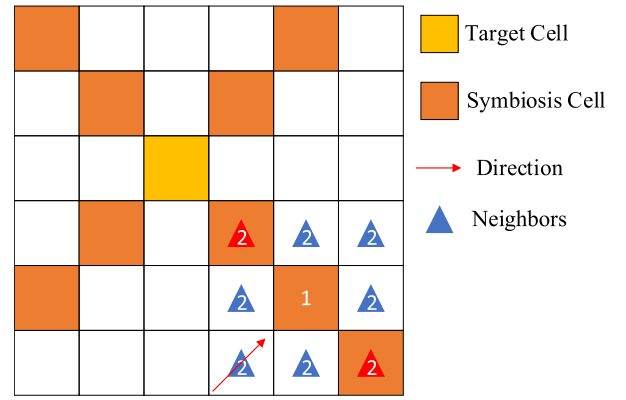


Fig. 4. Search process of the proposed method.

The search process of the sparse cell search method is shown in Fig. 4. For clarity, the grid cells whose detection statistics exceed the threshold are referred to as the potential target cells. The potential target cells include both the target cell and the symbiosis cells. When a potential target cell is detected (grid cell marked as 1), its adjacent cells (cells marked with triangles) will be added to the detection queue. Because the symbiosis cells are continuously distributed in space, more potential target cells will be detected (cells marked with red triangles). The adjacent cells of the newly detected potential target cells are continuously added to the detection queue, and this process is repeated until the detection queue is empty. Once any potential target cell is detected, this method can utilize the cell as a starting point to search all the potential target cells. It cleverly utilizes the symbiosis effect, significantly enhancing target detection efficiency.

The essence of this method is to reduce the number of cells that need to be searched and detected while ensuring that at least one symbiosis cell can be detected. We can select grid cells at fixed distance intervals in the CSR for detection. This method guarantees comprehensive coverage of the CSR and reduces the number of grid cells requiring processing. The interval can be flexibly adjusted according to factors such as SNR, the density of symbiosis cells, or specific detection requirements. Strategically selecting the grids to be processed instead of exhaustively searching for all grids. As a result, the detection process becomes more efficient, which can determine related cells faster and optimize resource allocation.

The steps of the sparse cell search method are as follows.

A. Sparse Cell Arrangement

The CSR is divided into G spatial grid cells. The size of grid is $\Delta x \times \Delta y$. The size of the CSR is $X \times Y$. The following relationship is established:

$$\begin{aligned} X &= n_x \cdot \Delta x \\ Y &= n_y \cdot \Delta y \end{aligned} \quad (20)$$

where the number of the grids is $G = n_x \cdot n_y$.

The center coordinate of any grid in the CSR can be expressed as

$$\begin{aligned} \mathbf{p}_g &= (x_i, y_j) = (x_0 + i \cdot \Delta x, y_0 + j \cdot \Delta y) \\ i &= 0, 1, 2, \dots, n_x - 1 \\ j &= 0, 1, 2, \dots, n_y - 1 \end{aligned} \quad (21)$$

where (x_0, y_0) is the center coordinate of the first grid in the lower left corner of CSR, Δx and Δy are the distances between the grids in the horizontal and vertical directions, respectively, and i and j are representing the grid cell index in the x -axis and y -axis directions, respectively.

After sparse grid arrangement, the number of sparse grids in the CSR becomes $1/n_{\text{times}}$ of the original. n_{times} is the sparse multiple, which can be flexibly adjusted according to factors such as SNR, the density of symbiosis cells, or specific detection requirements.

A two-fold sparse grid arrangement is taken as an example. The set of the sparse grids' center coordinates can be expressed as

$$\begin{aligned} \mathcal{Q} &= \left\{ (x_i, y_j) | i \bmod 2 = j \bmod 2, (x_i, y_j) \in \Gamma \right\} \\ i &= 0, 1, 2, \dots, n_x - 1 \\ j &= 0, 1, 2, \dots, n_y - 1. \end{aligned} \quad (22)$$

The computational complexity of the cell search method lies in the calculation of sample covariance matrix $\mathbf{C}_1, \dots, \mathbf{C}_N$ and the number of grids G . The computational complexity of the traversal cell search method is $\mathcal{O}(G \sum_{n=1}^N K_n M_n^2)$. The computational complexity of the sparse cell search method is $\mathcal{O}(\frac{G}{n_{\text{times}}} \sum_{n=1}^N K_n M_n^2)$. The computational complexity of the sparse cell search method is n_{times} -fold lower than the traversal cell search method.

B. Cell Search Sequence

After completing the sparse grid division of the CSR, the subsequent task is to determine the detection sequence. Our objective is to capture a symbiosis cell rapidly. The symbiosis cells are continuously distributed within the RRCs of different radar nodes covering the target. The radar's RRC is typically represented as a ring centered on the radar. When the radar's detection range extends, the distribution of the symbiosis cells within the CSR approximates several straight lines. Each line is approximately perpendicular to the line from the radar to the target. If the sparse grids are detected sequentially in the direction of the radar node aimed at the target, detection can occur most efficiently. However, we cannot get such a precise direction since the target's position remains unknown, and the multistatic radar system comprises multiple radar nodes.

A search direction can be determined according to the multistatic radar's geometrical configuration and the position of the CSR. Two typical deployment scenarios of multistatic radar systems are considered: linear deployment and circular deployment.

1) *Linear Deployment*: Radar nodes are deployed along a straight line or approximate straight line, as shown in Fig. 5(a). The direction from the midpoint of the multistatic radar's baseline toward the center of the CSR serves as the

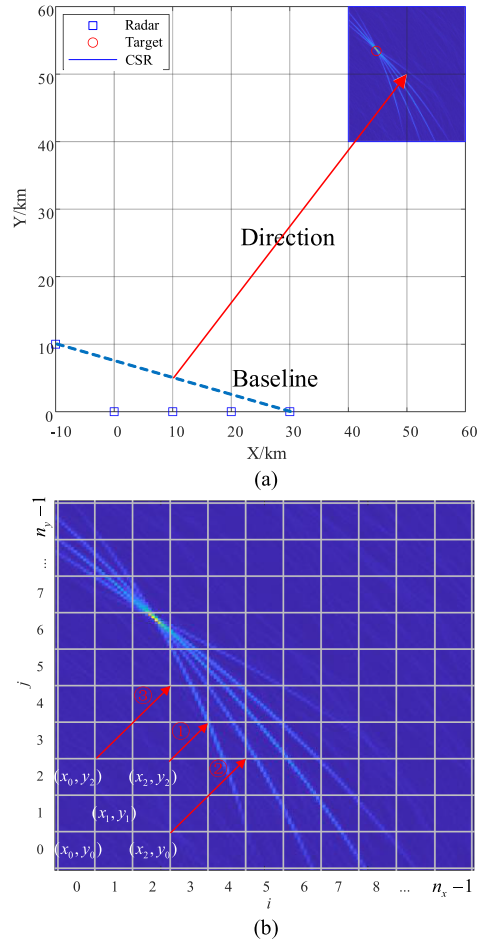


Fig. 5. Grid search sequence in a linear deployment scenario. (a) Linear deployment scenario. (b) Search sequence.

search direction

$$\mathbf{u} = \mathbf{p}_{\text{CSR}} - \sum_{n=1}^N \mathbf{p}_{R,n} / N \quad (23)$$

$$\theta_u = \arctan \left(\frac{y_u}{x_u} \right) \quad (24)$$

where $\mathbf{p}_{\text{CSR}} = (x_i^{\text{mid}}, y_j^{\text{mid}})$ is the center of the CSR.

In the 2-D scenario, since the CSR is discretized into multiple spatial grids, only eight standard directions are required to determine the next grid cell to be detected. The calculated search direction $\mathbf{u}$ can be mapped to one of eight standard directions (e.g., N, NE, E, SE, etc.) based on its angular orientation θ_u .

Once the search direction is determined, the detection process begins from the first grid (x_0, y_0) . For example, in the case of a 45° search direction, the sequence proceeds as $(x_0, y_0) \rightarrow (x_1, y_1) \rightarrow (x_2, y_2) \rightarrow \dots$ until the CSR boundary is reached. Subsequent starting points alternate between horizontal and vertical offsets. The second search starts at (x_2, y_0) , and the third search starts at (x_0, y_2) , as shown in Fig. 5(b). This alternating pattern continues until all sparse grid cells are covered. The rationale for this search sequence is to ensure that detection expands from the

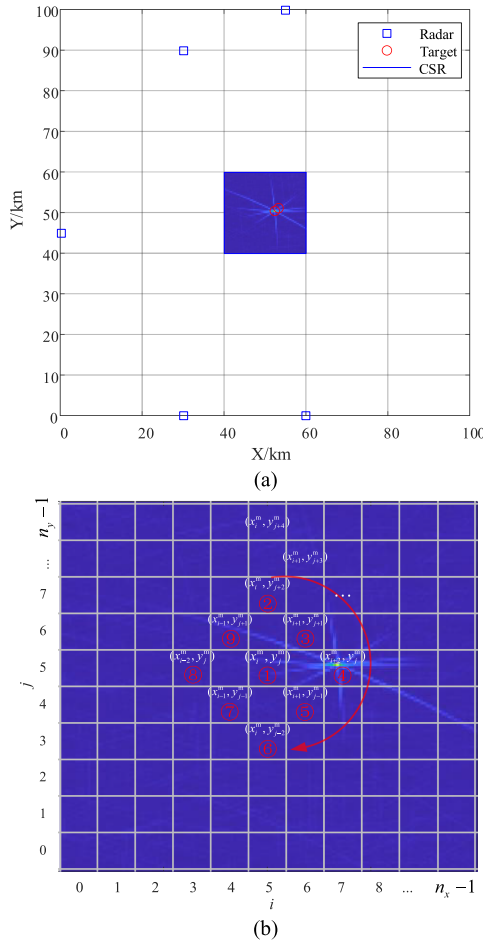


Fig. 6. Grid search sequence in a circular deployment scenario.
(a) Circular deployment scenario. (b) Search sequence.

central region of the CSR outward. Compared to edge-first detection strategies, prioritizing the central region improves efficiency. In practical applications, the starting point can be flexibly adjusted based on the search direction to maintain a center-to-edge detection order.

2) *Circular Deployment*: Radar nodes are deployed around the CSR in a circular or ringlike configuration. In this geometric configuration, the symbiosis cells are radially distributed in the CSR. Consequently, there is no need to calculate a specific search direction. Instead, detection can begin from the center of the CSR and proceed outward in concentric circular layers. For each layer expansion, the detection proceeds outward along eight standard directions. The search begins at the grid cell in the north direction and follows a clockwise sequence. The sequence proceeds as $(x_i^m, y_j^m) \rightarrow (x_{i+1}^m, y_{j+2}^m) \rightarrow (x_{i+1}^m, y_{j+1}^m) \rightarrow \dots$ until the CSR boundary is reached, as shown in Fig. 6. This approach can capture the symbiosis cells faster and reduce computational complexity.

C. Cell Search Algorithm

In the process of performing the search task, when a potential target cell is detected, a search algorithm is needed to locate the adjacent potential target cells. The cell search

algorithm can be seen in Algorithm 1, where p_g is the center coordinate of the initially detected potential target grid, s_g is the size of the grid, and e_d is the direction vector of the eight adjacent grids. In the 2-D rectangular coordinate system, e_d can be expressed as

$$e_d = \begin{bmatrix} -1 & 0 & 1 & 1 & 1 & 0 & -1 & -1 \\ 1 & 1 & 1 & 0 & -1 & -1 & -1 & 0 \end{bmatrix}. \quad (25)$$

When the detection statistic of a spatial grid exceeds the threshold, its center coordinate is input into the cell search algorithm. Subsequently, all adjacent grids of this potential target grid are placed into the set of grids to be detected Q_w . The grids in Q_w will be extracted and detected in turn. If the detection statistic of any grid in Q_w exceeds the detection threshold, the position of this grid and its detection statistic will be recorded in R . Similarly, the adjacent grids of this potential target grid will be placed in Q_w again. This detection process can be seen in lines 10 to 14 of Algorithm 1. During the execution of the algorithm, the grids that have been detected will be recorded in V to prevent repeated detections in subsequent processing, thus avoiding the waste of computational resources. The process is repeated until the detection queue Q_w is empty. Finally, all grids that exceed the threshold will be recorded in R for subsequent processing.

The essence of the cell search algorithm lies in its priority mechanism, which continuously expands the detection range to capture the symbiosis cells in the CSR. We no longer need to search all SRCs for detection and only need to ensure that at least one symbiosis cell is detected. This method reduces the computational burden and optimizes the target detection process, making centralized target detection more efficient and accurate.

Algorithm 1: Cell Search Algorithm.

Input: p_g, s_g, e_d
Output: R

- 1: Grids set for detection: $Q_w = \emptyset$;
- 2: Visited grids set: $V = \emptyset$;
- 3: Detection result set: $R = \emptyset$;
- 4: Obtain adjacent grids of p_g :
 $Q_w = p_g + s_g e_d = \{p_g^1, p_g^2, \dots, p_g^8\}$;
- 5: while $Q_w \neq \emptyset$ do
- 6: $p'_g = Q_w(1)$;
- 7: $Q_w(1) = \emptyset$;
- 8: if $V(p'_g) = \text{false}$
- 9: $V(p'_g) = \text{true}$;
- 10: if $\psi(p'_g) > \eta$ then
- 11: $R(p'_g) = \psi(p'_g)$;
- 12: $Q'_w = p'_g + s_g e_d$;
- 13: $Q_w = Q_w \cup Q'_w$;
- 14: end if
- 15: end if
- 16: end while
- 17: return R

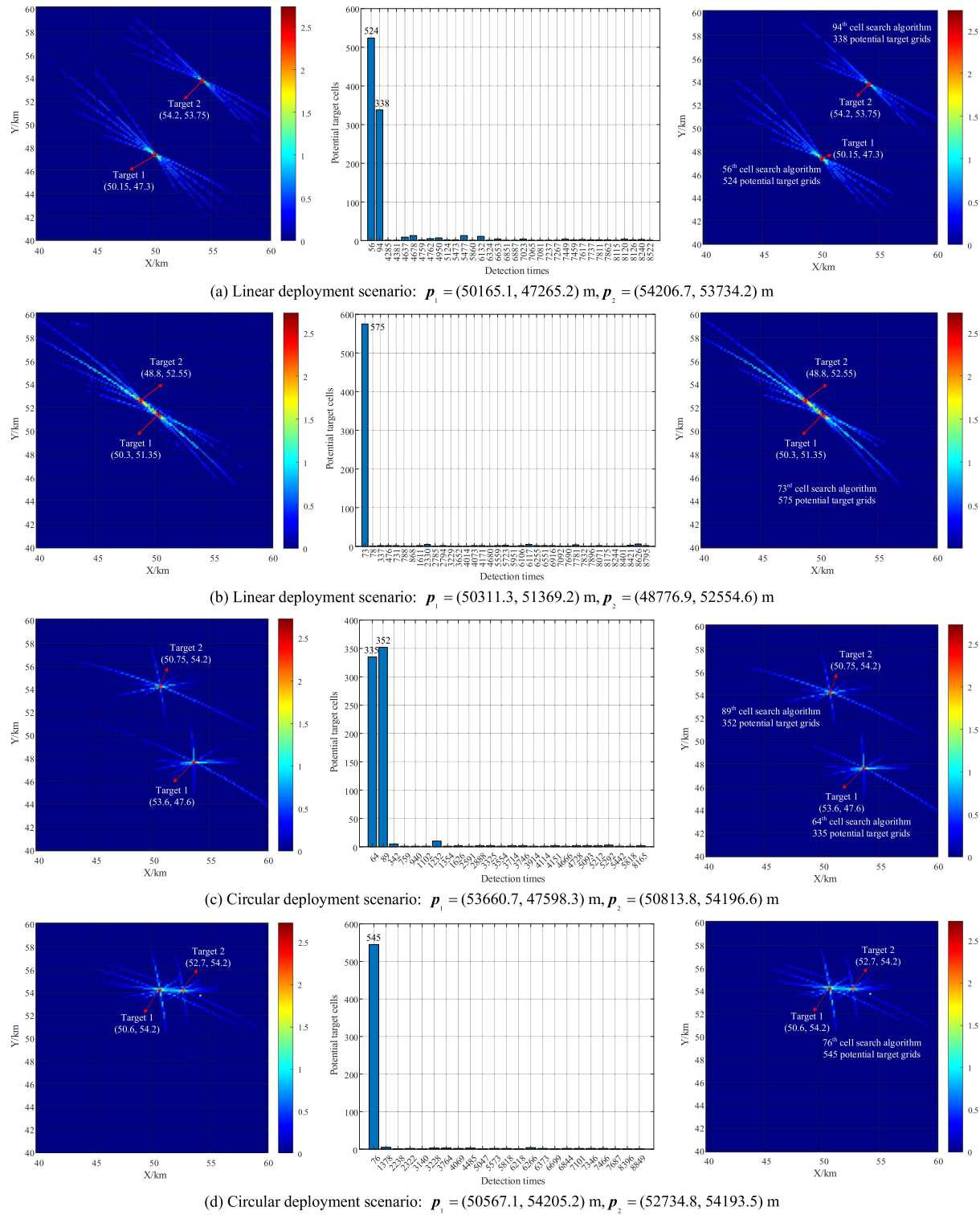


Fig. 7. Detection results of the sparse cell search method with two random targets under the linear and circular deployment scenarios.

IV. EXPERIMENT RESULTS

This section experimentally validates the sparse cell search method's detection performance and low computational complexity. Subsequently, we comprehensively analyze the method's performance under different SNRs. Furthermore, an alternating sparse cell search method is

proposed to handle the problem of discontinuous spatial distribution of symbiosis cells under low SNRs. Ultimately, the effectiveness of the alternating sparse cell search method is confirmed through comparative experiments with the traversal search method and detection experiments involving a moving target.

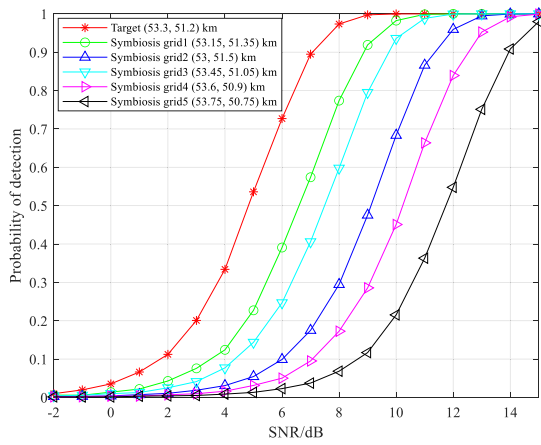


Fig. 8. Detection probability of the symbiosis grids with different SNR benchmark.

The linear deployment scenario is shown in Fig. 5(a). The radar nodes are located at $(-10, 10)$, $(0, 0)$, $(10, 0)$, $(20, 0)$, and $(30, 0)$ km. The circular deployment scenario is shown in Fig. 6(a). The radar nodes are located at $(0, 45)$, $(30, 0)$, $(60, 0)$, $(30, 90)$, and $(55, 100)$ km. In these two scenarios, all parameter settings remain the same except for the positions of the radar nodes. Considering the change of the radar cross-section of the target, the SNR weights of different radar nodes are set as 1.1, 0.8, 1.2, 1, and 1.1. The CSR of the multistatic radar is represented by a square area of $\Gamma = \{(x, y) \mid 40 \text{ km} \leq x \leq 60 \text{ km}, 40 \text{ km} \leq y \leq 60 \text{ km}\}$. All radars transmit linear frequency modulation signals with the bandwidth of 1 MHz. Therefore, the RRC of the radar node is $\frac{c}{2B_n} = 150 \text{ m}$. The sparse multiple is 2, and the set of the sparse grids is as (22). The search sequences of the sparse cells in linear deployment and circular deployment scenarios are shown in Figs. 5(b) and 6(b), respectively.

A. Performance Analysis of the Sparse Cell Search Method

To validate the robustness of the sparse cell search method, four comparative experiments are conducted: two under the linear deployment scenario and two under the circular deployment scenario. In the comparative experiments, two targets are randomly placed in the CSR. The targets' velocities are $(-150, -200)$ and $(-170, -190)$ m/s, respectively. The detection results of the sparse cell search method for the CSR are shown in Fig. 7. The left figure illustrates the overall detection results. The middle figure records the detection time for each individual potential target cell, as well as the total number of continuously distributed potential target cells. The detection results for clusters of continuously distributed potential target cells in space are shown in the right figure.

In experiments (a) and (b), the radar nodes are linearly deployed. In experiment (a), 524 continuously distributed potential target cells are detected at the 56th detection following the predefined search sequence. Subsequently, 338 continuously distributed potential target cells are detected at the 94th detection. In this experiment, the targets are far

apart, resulting in spatially separated clusters of cells, as shown in Fig. 7(a). The target positions can be determined by extracting the peak values from these two clusters. The center coordinates of the target cells are $(50.15, 47.3)$ and $(54.2, 53.75)$ km. In experiment (b), 575 continuously distributed potential target cells are detected at the 73rd detection. In this experiment, the targets are located close to each other, causing their symbiosis cells to overlap in space, as shown in Fig. 7(b). In this case, the target positions can be resolved using a geometric joint decision algorithm (GJDA) [23]. The center coordinates of the target cells are $(50.3, 51.35)$ and $(48.8, 52.55)$ km, respectively.

In experiments (c) and (d), the radar nodes are circularly deployed. In this geometric configuration, the symbiosis cells are radially distributed in the CSR. The detection begins from the center of the CSR and proceeds outwards in concentric circular layers. In experiment (c), 335 and 352 continuously distributed potential target cells are detected at the 64th and 89th detections, respectively. The target positions are determined by extracting the peak values from these two clusters, as shown in Fig. 7(c). The center coordinates of the target cells are $(53.6, 47.6)$ and $(50.75, 54.2)$ km, respectively. In experiment (d), 545 continuously distributed potential target cells are detected at the 76th detection. The symbiosis cells of the targets overlap in space, as shown in Fig. 7(d). The target positions are resolved using the GJDA. The center coordinates of the target cells are $(50.6, 54.2)$ and $(52.7, 54.2)$ km.

The two-fold sparse grids are used in the experiments. The traversal search method needs to detect 18 225 grids. The sparse cell search method only needs to detect 9113 grids. The computational complexity of the sparse cell search method is theoretically two times lower than the traversal search method. However, it is not necessary to detect all sparse grids to determine the target positions. The symbiosis cells are always closely related to the target cell. When the algorithm detects one or more clusters of continuously distributed cells in space, we have sufficient reason to conclude that the targets are located within these cells. Crucially, the sparse cell search algorithm immediately outputs detected clusters of continuously distributed potential target cells during the search process, enabling subsequent target localization. In contrast, traversal search method must complete the detection of all spatial grids before target localization. In experiments (a), (b), (c), and (d), the targets are successfully detected with 954, 647, 774, and 620 detections, which is significantly lower than the theoretical value of 9113. This characteristic enables the sparse cell search method to achieve target detection with lower computational complexity and higher efficiency compared to the traversal search method.

The sparse cell search method demonstrates excellent adaptability and scalability across different radar deployment scenarios and multitarget detection experiments. By reducing the number of detection cells and leveraging the symbiosis effect, the method exhibits highly efficient and stable detection performance with low computational complexity in multiple experiments. Experimental results

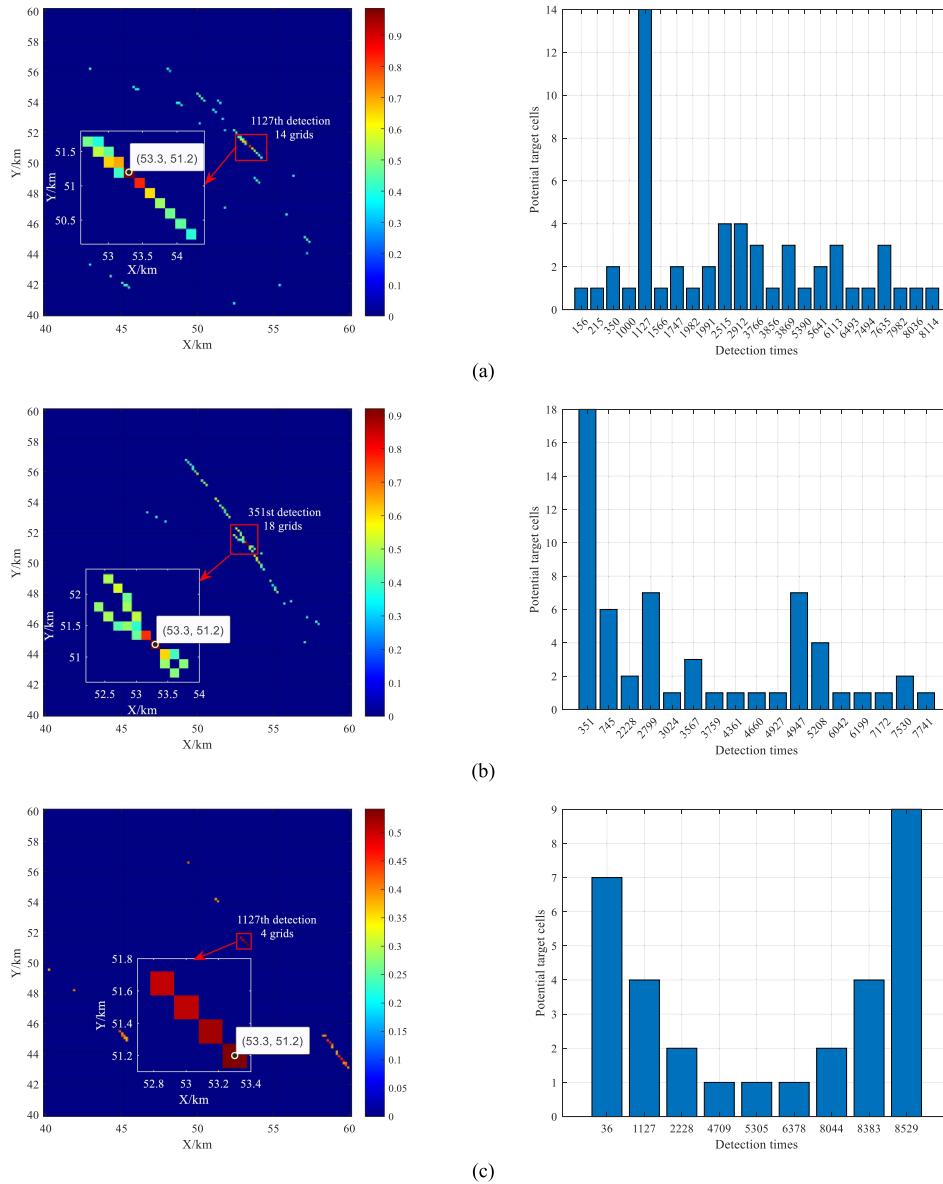


Fig. 9. Detection results of the sparse cell search method under different SNRs. (a) SNR = 12 dB (b) SNR = 9 dB (c) SNR = 8 dB

confirm that the sparse cell search method significantly reduces computational burden in practical applications while improving detection efficiency, making it has high application value.

B. Performance Analysis of the Sparse Cell Search Method Under Different SNR

This section explores the impact of SNR on the sparse cell search method. The radar nodes are linearly deployed. A target is placed at (53.25, 51.23) km. The center coordinate of the target grid is (53.3, 51.2) km. The center coordinates of the five symbiosis grids adjacent to the target grid are (53.15, 51.35), (53, 51.5), (53.45, 51.05), (53.6, 50.9), and (53.75, 50.75) km. Centralized target detection experiments are performed on these grids under different SNR benchmarks. Through 10^4 Monte Carlo experiments, the detection probability curves of these grids are shown in Fig. 8.

As the SNR decreases, the detection probabilities of both the target grid and the symbiosis grids gradually decrease. The detection probabilities of the symbiosis grids farther away from the target decrease even faster. The SNR decreases from 12 to 10 dB, resulting in the detection probabilities of the five symbiosis grids decreased by 1.8%, 6.4%, 27.7%, 38.9%, and 33.3%. When SNR is low, the distribution of symbiosis grids in space becomes discontinuous, adversely affecting the performance of the sparse cell search method. We must search more grids to find the symbiosis grid connected to the target. The simulation experiments under lower SNR benchmarks are carried out to analyze the detection performance of the sparse cell search method.

The position and velocity of the target are (53.3, 51.2) km and $(-150, -200)$ m/s. The SNR benchmarks of the multistatic radar are set to 12, 9, and 8 dB, respectively.

The detection results are displayed in Fig. 9. When the SNR is 12 dB, a symbiosis grid connected to the target is found after 1127 searches. There are 14 connected grids. Similarly, in a single target scenario, the grid with the largest detection statistics is the target grid, and (53.3, 51.2) km has been marked in the figure. When the SNR is 9 dB, a symbiosis grid connected to the target is found after 351 searches. There are 18 connected grids. When the SNR is 8 dB, a symbiosis grid connected to the target is found after 1127 searches. Only four grids are connected. When SNR is low, the distribution of symbiosis grids in space becomes discontinuous. More grids must be searched to find the target. By reasonably setting the direction and order of the search, the sparse cell search method can still significantly reduce the computational requirements, as shown in Fig. 9(a) and (b).

When the SNR is further reduced, the energy shared from the target is further reduced. The symbiosis cells can hardly be continuous in space. The false alarm caused by the noise is gradually increasing, and we can no longer rely on the number of continuously distributed symbiosis cells in space to estimate the region where the target is located, as shown in Fig. 9(c). It is necessary to detect all sparse grids to determine the target's position. However, if the target is located at the grid of the interval area, it will be missed. This article further proposes an alternating sparse cell search method to solve the problem of missed detection under low SNR.

C. Alternating Sparse Cell Search Method

When the SNR is low, the spatial distribution of symbiosis cells is no longer continuous. If the target is in the blind region of the sparse grid arrangement, there is a risk of missed detection. In this case, we cannot search for the target relying on a single symbiosis cell. Traversing all spatial grids in the CSR for detection can avoid missed detection. However, we strive to prevent this solution due to its inefficiency and the increased computational burden it entails. An alternating sparse cell search method is proposed in this article to solve the problem of missed detection under low SNR.

Continuous detection in the time domain can effectively make up for the blind spot of a single detection. In the scenario of two times sparsity, a complementary sparse grid arrangement method is designed, as shown in Fig. 10. The search and detection of the sparse grid are alternately performed during two coherent processing intervals. The alternating sparse cell search method can capture targets that may be undetectable in a single detection. In practical applications, moving targets may cross the blind region of the sparse grid many times. In this case, the method is particularly effective.

To explore the detection performance of the alternating sparse cell search method, a comparative experiment is conducted among the traversal search method, the sparse cell search method, and the alternating sparse cell search method. In the experiment, the radar nodes are linearly deployed, and a target is randomly placed in the CSR. The

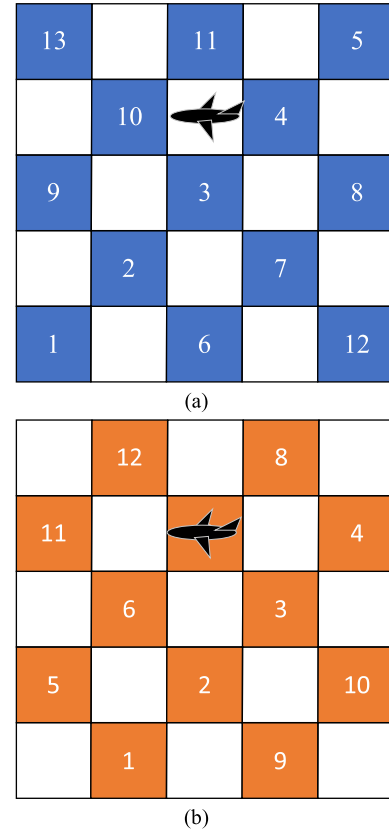


Fig. 10. Alternating sparse cell search method. (a) CPI 1. (b) CPI 2.

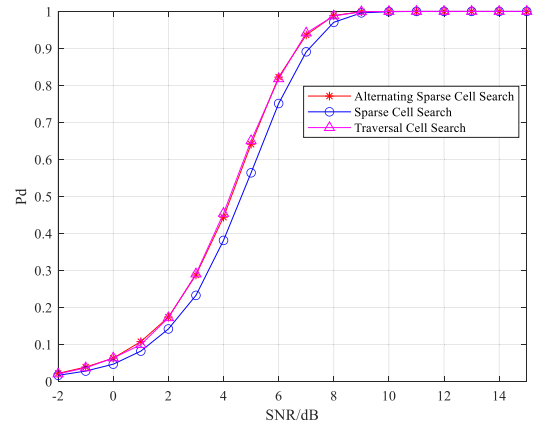


Fig. 11. Detection probability of the three search methods with different SNR benchmark.

target is detected under different SNR benchmarks. Through 10^4 Monte Carlo experiments, the detection probability curves of these three methods are shown in Fig. 11.

The alternating sparse cell search method has the same detection performance as the traversal search method, while its computational complexity is half that of the traversal search method. As demonstrated by the results of Experiment A, when the SNR is high, the alternating sparse cell search method exhibits even lower computational complexity and higher detection efficiency. Under low SNR conditions, the sparse cell search method is unable to handle missed detections effectively, resulting in lower detection

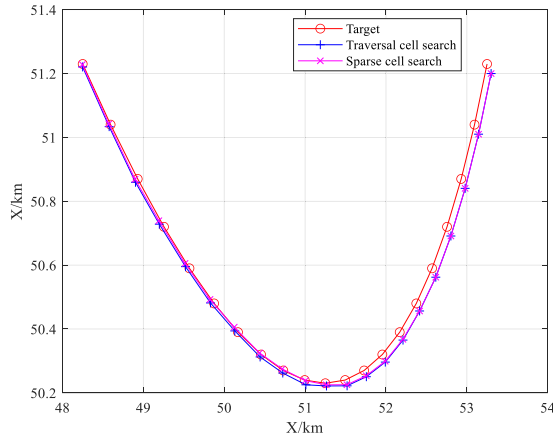


Fig. 12. Results of continuous observation.

performance compared to both the alternating sparse cell search method and the traversal search method. As the SNR increases, the detection performance of all three methods becomes comparable, which validates our analysis.

The alternating sparse cell search method is applied to continuous observation of a moving target to prove its effectiveness. A moving target is located at (53.25, 51.23) km. The initial velocity and acceleration of the target are $(-150, -200)$ m/s and $(-10, 20)$ m/s². Fig. 12 shows the results of continuous observation when the SNR = 8 dB.

The experimental results demonstrate that the proposed method has the same detection performance as the traversal search method in continuous observation scenarios, while its computational complexity is merely half that of the traversal search method. The alternating sparse cell search method is an improvement over the sparse cell search method, making it more suitable for practical detection scenarios. In conditions of high SNR, we can employ sparser grids to further reduce the computational complexity of cell search. This combination of high performance and low complexity improves the efficiency of cell search, so this method is especially suitable for multistatic radar search scenarios that require fast decision-making.

V. CONCLUSION

This article focuses on the cell search problem in SRC-based centralized target detection. In centralized target detection, the traversal cell search method suffers from high computational complexity, making it challenging to implement in practical engineering scenarios. This article proposes a sparse cell search method to solve the problem. The proposed method utilizes sparse SRCs to divide the detection space and preferentially captures the symbiosis cells. Subsequently, it utilizes the characteristic that these symbiosis cells are consistently closely related to the SRC containing the target to locate the real target. This method significantly reduces the computational complexity of the cell search. Simulation experiments demonstrate the proposed method's low computational complexity and robustness detection performance.

At low SNR, the spatial distribution of symbiosis cells becomes discontinuous, which adversely affects the performance of the sparse cell search method and may result in missed detections. To tackle this issue, we introduce an alternating sparse cell search method. This method reduces missed detection by multiple complementary detections. The experiments indicate that the proposed method achieves high detection performance while maintaining low computational complexity, making it suited for practical engineering applications.

APPENDIX

Minimize the denominator of (11) on $\alpha_1, \dots, \alpha_N$, and derivation of (20).

The denominator of (11) is the minimization of the following formula:

$$\min_{\alpha_1, \dots, \alpha_N} \prod_{n=1}^N \left(1 + \sum_{l=1}^{L_n} \mathbf{z}_{n,l}^H \mathbf{V}_n^{-1} \mathbf{z}_{n,l} - \alpha_n \sum_{l=1}^{L_n} s_{n,l}(\mathbf{p}) \mathbf{z}_{n,l}^H \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v}) - \alpha_n^* \sum_{l=1}^{L_n} s_{n,l}^*(\mathbf{p}) \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{z}_{n,l} + |\alpha_n|^2 \sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v}) \right)^{L_n + K_n}. \quad (26)$$

Since the signals received by different radar nodes are independent, the complex amplitudes of the received signals at each radar node can be estimated separately. Subsequently, (26) can be minimized utilizing $\hat{\alpha}_1, \dots, \hat{\alpha}_N$. The term $1 + \sum_{l=1}^{L_n} \mathbf{z}_{n,l}^H \mathbf{V}_n^{-1} \mathbf{z}_{n,l}$ is independent of α_n , and (26) can be further simplified as

$$\begin{aligned} & \min_{\alpha_1, \dots, \alpha_N} \prod_{n=1}^N \left(\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v}) \right. \\ & \quad \times \left(|\alpha_n|^2 - \alpha_n \frac{\sum_{l=1}^{L_n} s_{n,l}(\mathbf{p}) \mathbf{z}_{n,l}^H \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})}{\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})} \right. \\ & \quad \left. \left. - \alpha_n^* \frac{\sum_{l=1}^{L_n} s_{n,l}^*(\mathbf{p}) \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{z}_{n,l}}{\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})} \right) \right)^{L_n + K_n} \\ & = \min_{\alpha_1, \dots, \alpha_N} \prod_{n=1}^N \left(\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v}) \right. \\ & \quad \times \left(\alpha_n - \frac{\sum_{l=1}^{L_n} s_{n,l}^*(\mathbf{p}) \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{z}_{n,l}}{\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})} \right) \\ & \quad \times \left(\alpha_n^* - \frac{\sum_{l=1}^{L_n} s_{n,l}(\mathbf{p}) \mathbf{z}_{n,l}^H \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})}{\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})} \right) \\ & \quad \left. - \frac{\left(\sum_{l=1}^{L_n} s_{n,l}^*(\mathbf{p}) \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{z}_{n,l} \right) \left(\sum_{l=1}^{L_n} s_{n,l}(\mathbf{p}) \mathbf{z}_{n,l}^H \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v}) \right)}{\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})} \right)^{L_n + K_n}. \quad (27) \end{aligned}$$

Finally, (27) can be simplified as

$$\min_{\alpha_1, \dots, \alpha_N} \prod_{n=1}^N \left(\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v}) \right. \\ \times \left| \alpha_n - \frac{\sum_{l=1}^{L_n} s_{n,l}^*(\mathbf{p}) \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{z}_{n,l}}{\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})} \right|^2 \\ \left. - \frac{\left| \sum_{l=1}^{L_n} s_{n,l}^*(\mathbf{p}) \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{z}_{n,l} \right|^2}{\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})} \right)^{L_n + K_n}. \quad (28)$$

Obviously, when

$$\alpha_n = \frac{\sum_{l=1}^{L_n} s_{n,l}^*(\mathbf{p}) \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{z}_{n,l}}{\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})} \quad (29)$$

the original likelihood ratio function reaches its maximum value.

Substituting (29) into (11), we can get the statistic as

$$\max_{\mathbf{p}, \mathbf{v}} \prod_{n=1}^N \left(1 - \frac{q_n(\mathbf{p}, \mathbf{v})}{p_n} \right)^{-(L_n + K_n)} \underset{H_0}{\overset{H_1}{\geq}} \eta \quad (30)$$

where

$$p_n = 1 + \sum_{l=1}^{L_n} \mathbf{z}_{n,l}^H \mathbf{V}_n^{-1} \mathbf{z}_{n,l} \quad (31)$$

$$q_n(\mathbf{p}, \mathbf{v}) = \frac{\left| \sum_{l=1}^{L_n} s_{n,l}^*(\mathbf{p}) \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{z}_{n,l} \right|^2}{\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^H(\mathbf{p}, \mathbf{v}) \mathbf{V}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})}. \quad (32)$$

REFERENCES

- [1] S. H. Javadi and A. Farina, "Radar networks: A review of features and challenges," *Inf. Fusion*, vol. 61, pp. 48–55, 2020.
- [2] S. Zhang, Y. Zhou, L. Zhang, Q. Zhang, and L. Du, "Target detection for multistatic radar in the presence of deception jamming," *IEEE Sensors J.*, vol. 21, no. 6, pp. 8130–8141, Mar. 2021.
- [3] S. Shao, W.-Q. Wang, Y. Sun, and S. Zhang, "Effects of spatial position on mainlobe interference suppression performance of FDA-MIMO radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 6, pp. 9437–9443, Dec. 2024, doi: [10.1109/TAES.2024.3443778](https://doi.org/10.1109/TAES.2024.3443778).
- [4] T. Yuan, F. He, Z. Dong, Y. Su, Z. Fan, and L. Yu, "Suppress mainlobe deceptive jamming target under unambiguous range compensation based on FDA-MIMO radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 5, pp. 6853–6868, Oct. 2024, doi: [10.1109/TAES.2024.3407729](https://doi.org/10.1109/TAES.2024.3407729).
- [5] H. Liu, S. Zhou, H. Su, and Y. Yu, "Detection performance of spatial frequency diversity MIMO radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 50, no. 4, pp. 3137–3155, Oct. 2014.
- [6] J. Liu and J. Li, "Robust detection in MIMO radar with steering vector mismatches," *IEEE Trans. Signal Process.*, vol. 67, no. 20, pp. 5270–5280, Oct. 2019, doi: [10.1109/TSP.2019.2939078](https://doi.org/10.1109/TSP.2019.2939078).
- [7] M. Sadeghi, F. Behnia, R. Amiri, and A. Farina, "Target localization geometry gain in distributed MIMO radar," *IEEE Trans. Signal Process.*, vol. 69, pp. 1642–1652, Feb. 2021.
- [8] Y. Yang, H. Su, Q. Hu, S. Zhou, and J. Huang, "Spatial resolution cell based centralized target detection in multistatic radar," *Signal Process.*, vol. 152, pp. 238–246, Nov. 2018, doi: [10.1016/j.sigpro.2018.06.003](https://doi.org/10.1016/j.sigpro.2018.06.003).
- [9] Z. Zhang, Y. Zhou, S. Zhang, S. Shang, T. You, and L. Zhang, "Elliptical spatial resolution cell-based centralized target detection in multistatic radar," *IEEE Sensors J.*, vol. 24, no. 9, pp. 15367–15377, May 2024, doi: [10.1109/JSEN.2024.3369943](https://doi.org/10.1109/JSEN.2024.3369943).
- [10] J. Guo and H. Tao, "Cramer–Rao lower bounds of target positioning estimate in netted radar system," *Digit. Signal Process.*, vol. 118, 2021, Art. no. 103222.
- [11] R. Hoffmann, N. Neuberger, and R. Vehmas, "Rx beamforming for long baseline multistatic radar networks," in *Proc. IEEE Radar Conf.*, 2021, pp. 1–6.
- [12] Z. Yu, J. Li, Q. Guo, and J. Ding, "Efficient direct target localization for distributed MIMO radar with expectation propagation and belief propagation," *IEEE Trans. Signal Process.*, vol. 69, pp. 4055–4068, Jun. 2021.
- [13] Y. Wang, "Multi-sensor fusion tracking algorithm based on augmented reality system," *IEEE Sensors J.*, vol. 21, no. 22, pp. 25010–25017, Nov. 2021.
- [14] D. Dash and V. Jayaraman, "A probabilistic model for sensor fusion using range-only measurements in multistatic radar," *IEEE Sensors Lett.*, vol. 4, no. 6, Jun. 2020, Art. no. 7500604.
- [15] S. G. Dontamsetti and R. V. R. Kumar, "A distributed MIMO radar with joint optimal transmit and receive signal combining," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 1, pp. 623–635, Feb. 2021.
- [16] C. M. T. and T. A. J., "Relative entropy and mutual information," in *Elements of Information Theory*, 1st ed. New York, NY, USA: Wiley, 2005, pp. 12–21.
- [17] X. Xiao, S. Li, S. Peng, X. Xue, X. Zheng, and B. Zhou, "Microwave photonic wideband distributed coherent aperture radar with high robustness to time synchronization error," *J. Lightw. Technol.*, vol. 39, no. 2, pp. 347–356, Jan. 2021.
- [18] S. Chao, B. Chen, and C. Li, "Grid cell based detection strategy for MIMO radar with widely separated subarrays," *AEU - Int. J. Electron. Commun.*, vol. 66, no. 9, pp. 741–751, Sep. 2012, doi: [10.1016/j.aeue.2012.01.006](https://doi.org/10.1016/j.aeue.2012.01.006).
- [19] Q. Hu, H. Su, S. Zhou, Z. Liu, and J. Liu, "Target detection in distributed MIMO radar with registration errors," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 52, no. 1, pp. 438–450, Feb. 2016, doi: [10.1109/TAES.2015.140479](https://doi.org/10.1109/TAES.2015.140479).
- [20] Y. Yang, H. Su, Q. Hu, S. Zhou, and J. Huang, "Centralized adaptive CFAR detection with registration errors in multistatic radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 54, no. 5, pp. 2370–2382, Oct. 2018, doi: [10.1109/TAES.2018.2816467](https://doi.org/10.1109/TAES.2018.2816467).
- [21] L. Chen, T. Zhou, W. Yi, S. Li, L. Kong, and X. Yang, "Grid space searching based two-steps detection procedure for MIMO radar with widely separated antennas," in *Proc. IEEE 20th Int. Conf. Inf. Fusion*, Jul. 2017, pp. 1–5, doi: [10.23919/ICIF.2017.8009827](https://doi.org/10.23919/ICIF.2017.8009827).
- [22] W. Yi, T. Zhou, Y. Ai, and R. S. Blum, "Suboptimal low complexity joint multi-target detection and localization for non-coherent MIMO radar with widely separated antennas," *IEEE Trans. Signal Process.*, vol. 68, pp. 901–916, Jan. 2020, doi: [10.1109/TSP.2020.2968282](https://doi.org/10.1109/TSP.2020.2968282).
- [23] S. Yang, W. Yi, and A. Jakobsson, "Multitarget detection strategy for distributed MIMO radar with widely separated antennas," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, May 2022, Art. no. 5113516, doi: [10.1109/TGRS.2022.3175046](https://doi.org/10.1109/TGRS.2022.3175046).
- [24] Q. He, R. S. Blum, H. Godrich, and A. M. Haimovich, "Target velocity estimation and antenna placement for MIMO radar with widely separated antennas," *IEEE J. Sel. Top. Signal Process.*, vol. 4, no. 1, pp. 79–100, Feb. 2010.
- [25] Q. He, N. H. Lehmann, R. S. Blum, and A. M. Haimovich, "MIMO radar moving target detection in homogeneous clutter," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 46, no. 3, pp. 1290–1301, Jul. 2010.
- [26] P. Chen, L. Zheng, X. Wang, H. Li, and L. Wu, "Moving target detection using colocated MIMO radar on multiple distributed moving platforms," *IEEE Trans. Signal Process.*, vol. 65, no. 17, pp. 4670–4683, Sep. 2017.
- [27] H. Li, Z. Wang, J. Liu, and B. Himed, "Moving target detection in distributed MIMO radar on moving platforms," *IEEE J. Sel. Top. Signal Process.*, vol. 9, no. 8, pp. 1524–1535, Dec. 2015.

- [28] W. Liu, J. Liu, Y. Gao, G. Wang, and Y.-L. Wang, "Multichannel signal detection in interference and noise when signal mismatch happens," *Signal Process.*, vol. 166, 2020, Art. no. 107268.
- [29] J. Liu, W. Liu, X. Chen, D. Orlando, and A. Farina, "Performance analysis of the generalized likelihood ratio test in general phased array radar configuration," *IEEE Trans. Signal Process.*, vol. 69, pp. 4544–4555, Jul. 2021, doi: [10.1109/TSP.2021.3097658](https://doi.org/10.1109/TSP.2021.3097658).
- [30] W. Liu, C. Huang, Q. Du, B. Zhou, Z. Zhang, and H. Chen, "GLRT-based adaptive detector for colocated MIMO radar in partially homogeneous environment," in *Proc. 5th Int. Conf. Artif. Intell. Electron. Eng.*, Jan. 2024, pp. 65–68, doi: [10.1145/3658835.3658844](https://doi.org/10.1145/3658835.3658844).
- [31] P. Stoica, J. Li, and Y. Xie, "On probing signal design for MIMO radar," *IEEE Trans. Signal Process.*, vol. 55, no. 8, pp. 4151–4161, Aug. 2007, doi: [10.1109/TSP.2007.894398](https://doi.org/10.1109/TSP.2007.894398).
- [32] A. Aubry, A. D. Maio, S. Marano, and M. Rosamilia, "Single-pulse simultaneous target detection and angle estimation in a multi-channel phased array radar," *IEEE Trans. Signal Process.*, vol. 68, pp. 6649–6664, Nov. 2020, doi: [10.1109/TSP.2020.3039883](https://doi.org/10.1109/TSP.2020.3039883).
- [33] L. Lan et al., "GLRT-based adaptive target detection in FDA-MIMO radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 1, pp. 597–613, Feb. 2021, doi: [10.1109/TAES.2020.3028485](https://doi.org/10.1109/TAES.2020.3028485).
- [34] E. Kelly, "An adaptive detection algorithm," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-22, no. 2, pp. 115–127, Mar. 1986.



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